



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to M/s KP SANTHAN ENTERPRISES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

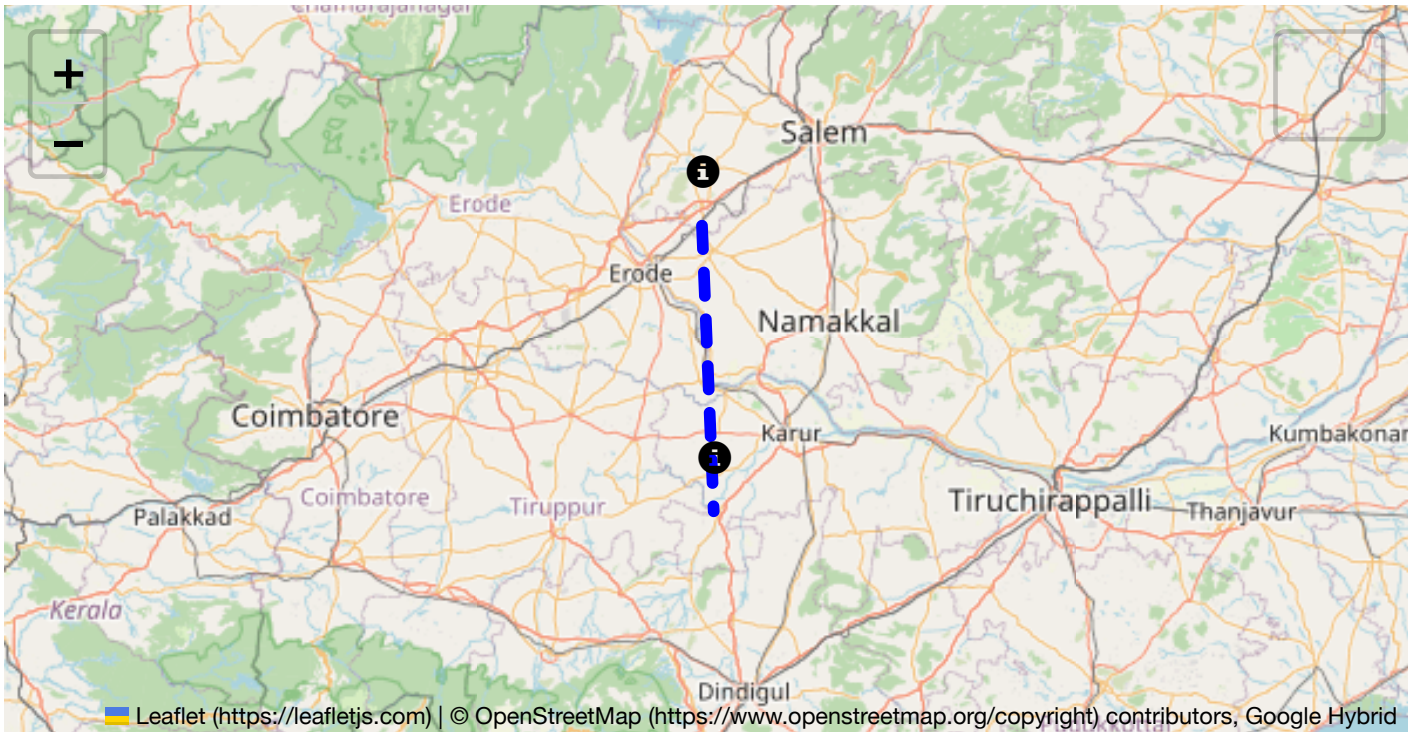
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.

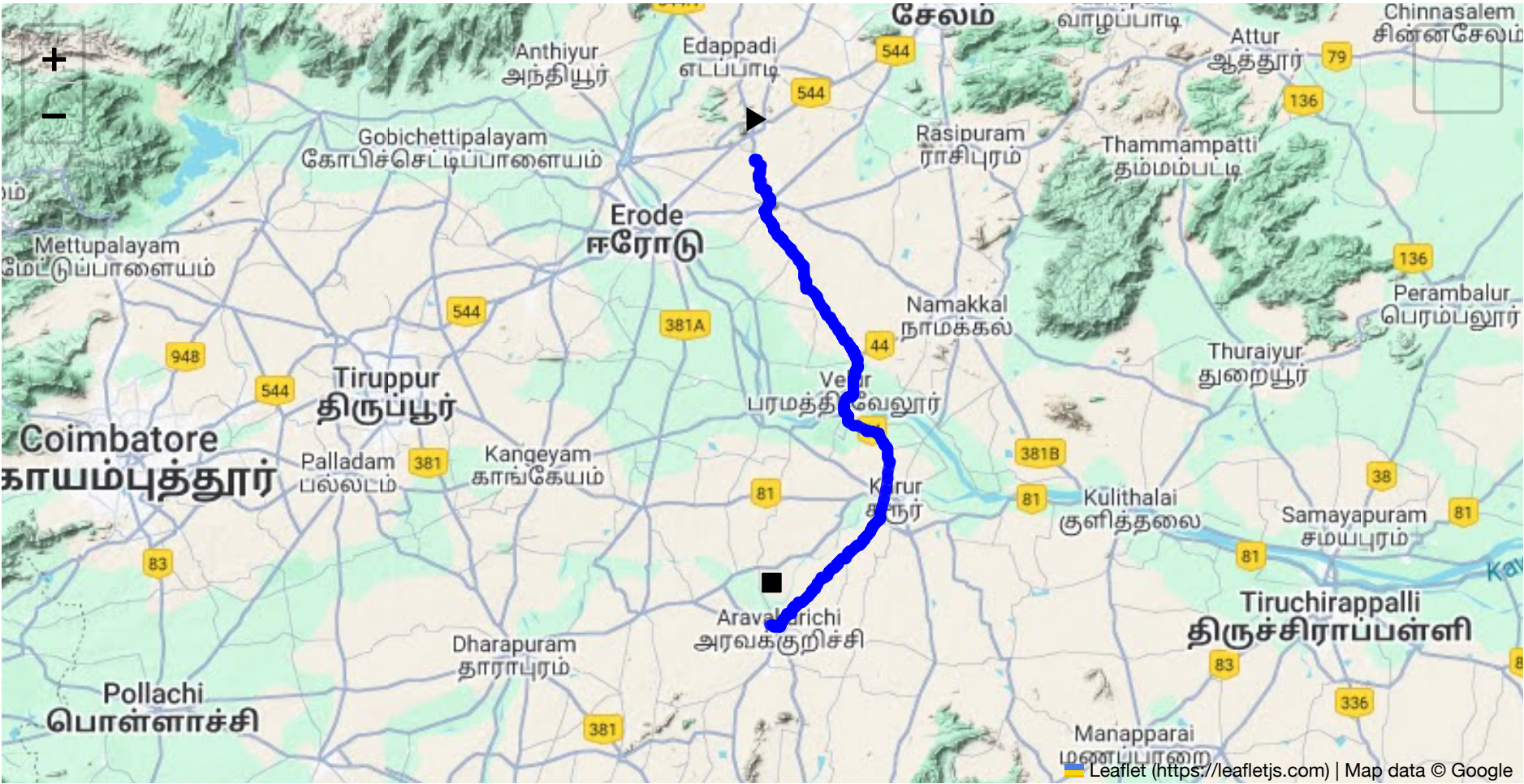


Route Summary:
Total Distance: 93.61 km
Estimated Duration: 1.7 hours

Adjusted Duration (Heavy Vehicle): 2.2 hours

Start: (11.4381, 77.8734)

End: (10.779832, 77.896105)



Welcome to the Journey Risk Management Study

Route Overview

The journey from CVQF+23W, Sangagiri, to QVJV+4W8, Arikkaranvalasu, covers approximately 93.61 kilometers and generally takes about 1.74 hours by heavy vehicle, specifically those transporting hazardous materials. The route typically traverses state highways and rural roads which pass through a mix of urban, semi-urban, and rural villages and towns in Tamil Nadu.

Weather Conditions

Tamil Nadu experiences a tropical climate. During the southwest monsoon (June to September), heavy rainfall can occur, causing waterlogging and affecting road conditions. The dry season runs from January to May, where high temperatures and occasional dust storms might impair visibility. The northeast monsoon (October to December) can also bring rain. Sudden weather changes necessitate attentiveness, especially during monsoon months.

Traffic Patterns

Traffic may be moderate to heavy, particularly near urban areas and market towns. Peak traffic hours align with standard Indian business hours, from 8 AM to 10 AM and 5 PM to 7 PM. Key congestion-prone areas include market streets in busy towns and junctions that lead into larger highways or significant local roads.

Road Quality and Infrastructure

State highways in Tamil Nadu are generally well-maintained, though some rural connectors might have potholes or uneven pavements. Be prepared for occasional narrow passages, particularly through small villages. Ensure adherence to posted speed limits due to variable road conditions.

Alternative Routes

In emergencies, routes through NH44 might offer faster access to major urban centers with better facilities. Local roads can provide access to intermediate points, though they may be less direct and variable in quality.

Local Regulations on Hazardous Materials

Tamil Nadu mandates strict guidelines for the transport of hazardous materials. Drivers should hold appropriate licenses, ensure that vehicles are correctly marked, and adhere to specific routing restrictions in populated or environmentally sensitive areas. Compliance with these is crucial for legal and safety reasons.

Historical Incidents

The region has historically seen limited major incidents involving heavy vehicles due to its adherence to strict regulations and relatively developed infrastructure. However, isolated cases have occurred due to weather-induced accidents or mechanical failure.

Environmental Considerations

Drivers should be mindful of reserved forests and water bodies along the route. Adherence to rules on pollution control and waste management is vital to avoid fines and protect the environment.

Communication Coverage

Most of the route benefits from good mobile coverage, though certain rural sections could experience connectivity issues. Drivers are advised to test communication equipment before and during the journey.

Emergency Response Times

Estimated emergency response times might range from 30 minutes to over an hour, depending on proximity to urban centers and availability of services. Areas near major towns will likely have faster response times.

Overall Summary of Risk Assessment

The route from CVQF+23W, Sangagiri, to QVJV+4W8, Arikkaranvalasu, while generally safe for heavy vehicles, involves certain risks from weather conditions, traffic during peak hours, and rural road quality issues. Following local driving regulations, maintaining communication, and being prepared for variable weather conditions can mitigate many of these risks. Continuous monitoring and strategic planning can enhance safety and efficiency throughout the journey.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44019, 77.87547	15 KM/Hr
3	Turn	Medium	11.39685, 77.88861	30 KM/Hr
4	Turn	High	11.37862, 77.89488	15 KM/Hr
5	Turn	High	11.37850, 77.89453	15 KM/Hr
6	Turn	Medium	11.16764, 78.01565	30 KM/Hr
7	Turn	High	11.14545, 78.02208	15 KM/Hr
8	Turn	High	11.14480, 78.02291	15 KM/Hr
9	Turn	Medium	11.10102, 78.00572	30 KM/Hr
10	Turn	High	10.79637, 77.92684	15 KM/Hr
11	Turn	High	10.77656, 77.91017	15 KM/Hr
12	Turn	Medium	10.77761, 77.90150	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium
18	college	valluvar catering	10.9553473, 78.0606583	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44019, 77.87547



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39685, 77.88861



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37862, 77.89488



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37850, 77.89453



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.16764, 78.01565



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14545, 78.02208



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14480, 78.02291



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.10102, 78.00572



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.79637, 77.92684



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.77656, 77.91017



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.77761, 77.90150

